Allen Hsieh

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Education

University of California, Berkeley Expected May 2030
B.S. Aerospace Engineering, College of Engineering · Berkeley, CA

Troy High School May 2026
Troy Tech Magnet Program · Fullerton, CA

Experience

Student Lab Assistant Oct 2026 – Present
Fisher Lab, UC Berkeley · Berkeley, CA

- Maintain fly stocks and provide engineering support (3D printing, microcontrollers, rig building) for a neuroscience lab studying how fruit fly compass neurons recalibrate from visual input.

Logistics Director Fall 2026 – Present
Hackathons @ Berkeley · Berkeley, CA

- Plan catering and hacker buses for Cal Hacks 13.0 (3,000+ attendees); source its merchandise from quote to delivery.

Avionics Team Member Fall 2026 – Present
Cal Aero SAE · Berkeley, CA

- Built a universal motor test jig that measures thrust to choose motors for the competition RC plane.
- Designing an ESP32 telemetry and black-box system for autonomous turbulence correction via propeller power.

Hacker & Attendee Summer 2026
Y Combinator Startup School · San Francisco, CA

- Attended YC's in-person Startup School; placed 5th of 1,000+ at the Corgi Hackathon with RunItBack, built in 12 hours, which turns one photo into a 3D environment for dementia reminiscence therapy.

Systems Engineering Intern Summer 2025
Raytheon · Fullerton, CA

- Reverse-engineered an Arduino-based robotic system and produced its technical documentation, wiring diagrams, and CAD models.
- Shadowed engineering workflows across the mechanical, electrical, software, and systems departments; attended product development briefings and radar test site visits.

Engineer Intern Jun – Aug 2024
Air Force Research Laboratory · Albuquerque, NM

- Workshop and museum assistant at Explora Science Center and Kirtland AFB; built prototypes with FDM and resin 3D printing, laser cutting, and lathe and mill CNC machining.
- Personal project: a staged-recovery rocket, taken through 11 design iterations to a 3,200 ft flight and a parachute recovery.

Leadership

Competitor & Windows Instructor Aug 2022 – May 2026
AFA CyberPatriot

- Four-year competitor as the team's Microsoft Windows OS specialist; 2x National Finals 2nd place (1,406 teams in CyberPatriot 17) and 2x Cisco Networking Challenge 1st place (2025, 2026).
- Taught middle and high school students as a Windows instructor for Troy High School's CyberPatriot program (2024 – 2026).

Co-Founder & Lead Programmer Jul 2023 – May 2026
Troy VEX Robotics

- Co-founded the school's first VEX team with 11 members in 2023; grew it to 7 teams and 40+ members in 2025 and 14 teams and 90+ in 2026; ran a robotics camp for 50+ students.
- Programmed autonomous navigation in C++ using sensor fusion, odometry, motion profiling, and Monte Carlo algorithms.

Founder & President Jul 2023 – May 2026
Troy Preparedness Club

- Founded a school-wide emergency preparedness initiative; ran 3 Teen CERT and 2 First Aid/BLS trainings with the fire department's emergency management branch, certifying about 150 students and reaching 300+ families.
- Facilitated 2,000+ volunteer hours; U.S. Congressional Certificate of Recognition; FEMA Youth Preparedness Council team lead (Region 9).

Founder & Team Captain Jul 2023 – May 2026
American Rocketry Challenge

- Founded and captained a competitive rocketry team; raised $2,000 from business sponsors.

Skills

Programming & tools	C++, Python, MATLAB, TypeScript, Next.js, React, Git, MySQL, LaTeX
Design & fabrication	SolidWorks (CAD and FEA), Fusion 360, OpenRocket, CNC machining, FDM and SLA printing
Certifications	GIAC GFACT (cybersecurity), SolidWorks CSWA, FEMA CERT Basic Training, Red Cross CPR/AED

Awards

- Corgi Hackathon, 5th of 1,000+, Y Combinator Startup School · Summer 2026
- CyberPatriot National Finals, 2nd place, and Cisco Networking Challenge, 1st place · 2025 and 2026
- VEX Robotics, 2x National Division Finalist, 4x VEX Worlds qualified, 3x California R4 States qualified
- NCSF Scholar with Honors · About the top 1% of 45,000 applicants, 2023
- National Merit Commended Scholar and AP Scholar with Distinction
- U.S. Congressional Certificate of Recognition, and VFW Citation and Medal